

The Periodic Table as ξ -Geometry:
Radial Mass Ladder and Angular Shell
Structure
from a Common Origin

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Chapter A

The Periodic Table as ξ -Geometry

Abstract

The periodic table of the elements and the particle-mass ladder of T0 theory share the same geometric quantum number n , but in different roles. Particle masses follow a *radial* exponential scaling $m_n \sim \xi^{-n} \cdot m_P$, while the shell structure of the periodic table obeys an *angular* quadratic degeneracy $g_n = 2n^2$. Both arise from the three-dimensional lattice geometry of T0 theory: ξ is the packing deficit of the 3D lattice (radial direction), $2n^2$ is the degeneracy of the n -th spherical shell in the same lattice (angular direction). The Rydberg energy lies exactly on rung 7 of the ξ -power ladder: $R_\infty = \frac{m_e \alpha^2}{2} \approx \xi^{6.956} \cdot E_P$. The duality principle $E_n \cdot g_n = 2R_\infty = \text{const}$ connects the $1/n^2$ energy ladder with the n^2 filling ladder and is a geometric equipartition. The periodic table is the angular cross-section of the ξ -geometry at rung $\xi^{\sim 7}$ — analogously, the lepton masses are the radial cross-section at rungs $\xi^{4.9}$ – $\xi^{5.8}$.

A.1 Introduction and Problem Statement

T0 theory (Time-Mass Duality / FFGFT) derives all physical constants from the geometric parameter

$$\xi = \frac{4}{30\,000} = 1.\bar{3} \times 10^{-4} \quad (\text{A.1})$$

[1, 3, 4]. The particle masses of the Standard Model organise themselves on an exponential ladder

$$m_n \sim \xi^{-n} \cdot m_P, \quad n = 1, 2, 3, \dots \quad (\text{A.2})$$

where n is the T0 quantum number and m_P is the Planck mass.

The periodic table of the elements, on the other hand, shows a very different pattern: shell capacities follow

$$g_n = 2n^2 = 2, 8, 18, 32, 50, \dots \quad (\text{A.3})$$

and binding energies scale as $1/n^2$. At first glance, particle masses and the periodic table appear unrelated.

Gemini's observation (2026): the periodic table lies "one level higher" than the particle masses — both have quantum numbers, but what common geometric origin connects their patterns?

The answer of this document: *T0 geometry has two axes — a radial one (ξ^n scaling) and an angular one ($2n^2$ degeneracy), both arising from the three-dimensional lattice structure with packing deficit ξ .*

A.2 The ξ -Power Ladder and Its Rungs

Previous documents in the T0 series have built up the ξ -power ladder step by step. Table A.1 shows the current state including the new atomic rung.

The Rydberg energy lies between rungs 6 and 7:

$$\frac{R_\infty}{E_P} = \frac{m_e}{m_P} \cdot \frac{\alpha^2}{2} = \xi^{5.775} \cdot \xi^{1.181} = \xi^{6.956} \approx \xi^7 \quad (\text{A.4})$$

The exponent is the sum of two independent T0 quantities: the electron mass (rung 5.775, Doc. 056) and the fine-structure constant (rung 1.181, Doc. 011).

Table A.1: Complete ξ -power ladder (as of March 2026)

Rung	ξ power	Physical content	Document
0	$\xi^0 = 1$	Planck scale, $L_0 = \xi \cdot \ell_P$	Doc. 009
1/2	$\xi^{1/2} \approx 10^{-2}$	$\alpha_W, m_e/m_\mu$, PPLN Δn	Doc. 041, 167
1	$\xi^1 \approx 10^{-4}$	$\alpha_{EM} \sim \xi \cdot E_0^2$	Doc. 011
4	$\xi^4 \approx 10^{-16}$	CMB temperature T_{CMB}	Doc. 166
23/4	$\xi^{5.775}$	Electron mass m_e/m_P	Doc. 056
~ 7	$\xi^{6.956}$	Rydberg energy R_∞, periodic table	this doc.
10	$\xi^{10} \approx 10^{-43}$	Hubble constant H_0	Doc. 165

A.3 Two Axes of ξ -Geometry

Radial Axis: Exponential Mass Ladder

Particle masses organise themselves along the *radial* axis of the T0 lattice. The fundamental relation is [1]:

$$m_n \sim \xi^{-n} \cdot m_P, \quad n = 1, 2, 3, \dots \quad (\text{A.5})$$

Each step on this ladder multiplies the mass by $\xi^{-1} = 7500$. The leptons occupy the rungs (Doc. 056, 122):

$$m_e : \xi^{5.775} \cdot m_P \quad (\text{A.6})$$

$$m_\mu : \xi^{5.177} \cdot m_P \quad (\text{A.7})$$

$$m_\tau : \xi^{4.861} \cdot m_P \quad (\text{A.8})$$

The mass ratio $m_e/m_\mu \propto \xi^{1/2}$ (Doc. 041) lies exactly on the electroweak exponent — the half-rung of the ξ -ladder.

Angular Axis: Quadratic Degeneracy Ladder

The *angular* axis of the same lattice determines the degeneracy of spherical shells. In 3D space, the n -th principal shell has:

$$g_n = 2 \sum_{l=0}^{n-1} (2l+1) = 2n^2 \quad (\text{A.9})$$

The factor 2 comes from spin $m_s = \pm \frac{1}{2}$; the sum is pure angular-momentum algebra in 3D space. Table A.2 shows the first four shells.

Table A.2: Shell structure and the periodic table

n	$g_n = 2n^2$	Period length	Noble gas Z	Period number
1	2	2	He (2)	1
2	8	8	Ne (10)	2, 3
3	18	18	Ar (18)	4, 5
4	32	32	Kr (36)	6, 7

The Connection through Quantum Number n

The same quantum number n appears in both systems, but with different functional roles:

$$\text{Radial (masses): } m_n \sim \xi^{-n} \cdot m_P \quad (\text{exponential}) \quad (\text{A.10})$$

$$\text{Angular (shells): } g_n = 2n^2 \quad (\text{quadratic}) \quad (\text{A.11})$$

This is not coincidental. In T0 lattice geometry, n corresponds to the *discrete radius in the lattice*:

- Radially: n steps outward from the lattice $\Rightarrow$ mass scales as ξ^{-n} (geometric sequence)
- Angularly: the lattice points on the sphere of radius n carry degeneracy $2n^2$ (combinatorial sequence)

A.4 The Duality Principle

Energy Ladder and Filling Ladder Are Dual

On the atomic scale (hydrogen-like atoms):

$$\text{Energy ladder: } E_n = -\frac{R_\infty}{n^2} \quad (\text{falls as } 1/n^2) \quad (\text{A.12})$$

$$\text{Filling ladder: } g_n = 2n^2 \quad (\text{grows as } n^2) \quad (\text{A.13})$$

Their product is exactly constant:

$$E_n \cdot g_n = \frac{R_\infty}{n^2} \cdot 2n^2 = 2R_\infty = \text{const} \quad (\text{A.14})$$

This is geometric *equipartition*: every shell contributes the same total energy $2R_\infty$ to the Rydberg scale, regardless of n . The $1/n^2$ energy ladder and the n^2 filling ladder are *inversely dual*.

Comparison with the Mass Ladder

Table A.3 summarises all three ladders.

Table A.3: Three ladders — one geometric origin

System	Ladder	Function of n	Type
Particle masses	$m_n \sim \xi^{-n} \cdot m_p$	exponential	radial
Shell energy	$E_n = R_\infty/n^2$	harmonic	angular
Shell filling	$g_n = 2n^2$	quadratic	angular
Duality: $E_n \cdot g_n = 2R_\infty = \text{const}$			

A.5 The Role of Fractal Dimension

In T0 (Doc. 009, 133):

$$D_f = 3 - \xi = 2.999867 \quad (\text{A.15})$$

Space dimension is almost, but not exactly, 3.

In the limit $D_f \rightarrow 3$ ($\xi \rightarrow 0$):

- $g_n = 2n^2$ is *exact* (pure spherical symmetry in the $\mathbb{Z}^3$ lattice)
- the shell structure of the periodic table is exact

For $D_f = 3 - \xi < 3$ there is a minimal correction:

$$\delta g_n \sim \xi \cdot n^2 \quad (\text{A.16})$$

For $n = 2$: $\delta g_2 \approx \xi \cdot 4 \approx 5 \times 10^{-4}$ (entirely negligible).

The periodic table is therefore exact because $D_f \approx 3$ to four decimal places. The tiny deviation $\xi \approx 1.3 \times 10^{-4}$ makes space “almost-integer-dimensional” — and precisely this deviation generates all the physics of particle masses and fundamental constants.

A.6 Numerical Verification

Rydberg Energy on the ξ -Ladder

$$R_\infty = \frac{m_e \alpha^2}{2} = 13.5984 \text{ eV} \quad (\text{A.17})$$

$$\frac{R_\infty}{E_p} = \frac{m_e}{m_p} \cdot \frac{\alpha^2}{2} = \xi^{5.775} \cdot \xi^{1.181} = \xi^{6.956} \quad (\text{A.18})$$

Table A.4: Rydberg energy: T0 derivation vs. measurement

Quantity	Value	ξ -exponent
m_e/m_p	4.185×10^{-23}	5.775
$\alpha^2/2$	2.663×10^{-5}	1.181
R_∞/E_p	1.114×10^{-27}	6.956
R_∞ (T0, approx.)	$\approx 9.1 \text{ eV } (\xi^7)$	7.000
R_∞ (measured)	13.598 eV	6.956

Duality Relation: Numerical Verification

$$E_1 \cdot g_1 = 13.60 \text{ eV} \cdot 2 = 27.20 \text{ eV} = 2R_\infty \quad (\text{A.19})$$

$$E_2 \cdot g_2 = 3.40 \text{ eV} \cdot 8 = 27.20 \text{ eV} = 2R_\infty \quad (\text{A.20})$$

$$E_3 \cdot g_3 = 1.51 \text{ eV} \cdot 18 = 27.18 \text{ eV} \approx 2R_\infty \quad (\text{A.21})$$

$$E_4 \cdot g_4 = 0.85 \text{ eV} \cdot 32 = 27.20 \text{ eV} = 2R_\infty \quad (\text{A.22})$$

The duality relation (A.14) holds exactly for all n .

A.7 The Periodic Table in the T0 Hierarchy

ξ -rung	Physics	Axis
$\xi^0 = 1$	Planck scale ($L_0 = \xi \cdot \ell_p$)	—
$\xi^{1/2}$	Electroweak: $\alpha_W, m_e/m_\mu$	radial
ξ^1	α_{EM} , EM coupling	radial
ξ^4	CMB temperature T_{CMB}	radial
$\xi^{5.8}$	Particle masses (leptons, quarks)	radial
$\xi^{6.956}$	Rydberg R_∞, periodic table	both
ξ^{10}	Hubble constant H_0	radial

Figure A.1: Position of the periodic table in the complete ξ -hierarchy

The special feature of the atomic rung ξ^7 : here *both* axes act. The radial axis determines the energy scale R_∞ ; the angular axis determines the pattern $2n^2$. The periodic table is the visible result of both axes acting together.

A.8 Predictions and Open Questions

Angular Corrections from $D_f < 3$

The fractal dimension $D_f = 3 - \xi$ leads to minimal corrections of the exact $2n^2$ rule:

$$g_n^{(T0)} = 2n^2 \cdot (1 - \xi \cdot f(n)) \quad (\text{A.23})$$

where $f(n)$ is a dimensionless function of n (still to be derived). For $n \leq 4$ the correction is $< 10^{-3}$ and not measurable.

Molecular Binding Energies

Chemical binding energies lie in the range 1–10 eV, hence also on rung ξ^7 . It remains to be checked whether specific binding energies lie on rational multiples of $\xi^{7/2} \cdot R_\infty$.

Heavy Elements and Relativistic Corrections

For $Z \gg 1$ relativistic effects become important. In T0 these are connected with the $\xi^{1/2}$ electroweak rung. A systematic analysis of ionisation energies of heavy elements ($Z > 54$) might reveal ξ -matches at the half-rung $\xi^{7-1/2} = \xi^{13/2}$.

A.9 Summary

1. **Energy scale:** $R_\infty/E_p = \xi^{6.956} \approx \xi^7$. The Rydberg energy — and hence all of atomic physics — lies on rung 7 of the ξ -power ladder.
2. **Radial axis:** Particle masses follow $m_n \sim \xi^{-n} \cdot m_p$. The exponent n is the discrete lattice radius in the radial ξ -direction.
3. **Angular axis:** Shell capacities follow $g_n = 2n^2$. The index n is the same discrete lattice radius, now acting as a shell index in the angular 3D-sphere direction.
4. **Duality principle:** $E_n \cdot g_n = 2R_\infty = \text{const.}$ The $1/n^2$ energy ladder and the n^2 filling ladder are inversely dual. Every shell contributes the same total energy $2R_\infty$ to the ξ^7 scale.

5. **Fractal dimension:** $D_f = 3 - \xi \approx 3$ makes $2n^2$ exact (to $\sim 10^{-4}$). The periodic table is exact because space is almost-integer-dimensional.
6. **Common origin:** Particle masses and the periodic table arise from the same 3D lattice geometry with packing deficit ξ . The periodic table is the *angular cross-section* of ξ -geometry at rung $\xi^{\sim 7}$; the lepton masses are the *radial cross-section* at rungs $\xi^{4.9} - \xi^{5.8}$.

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